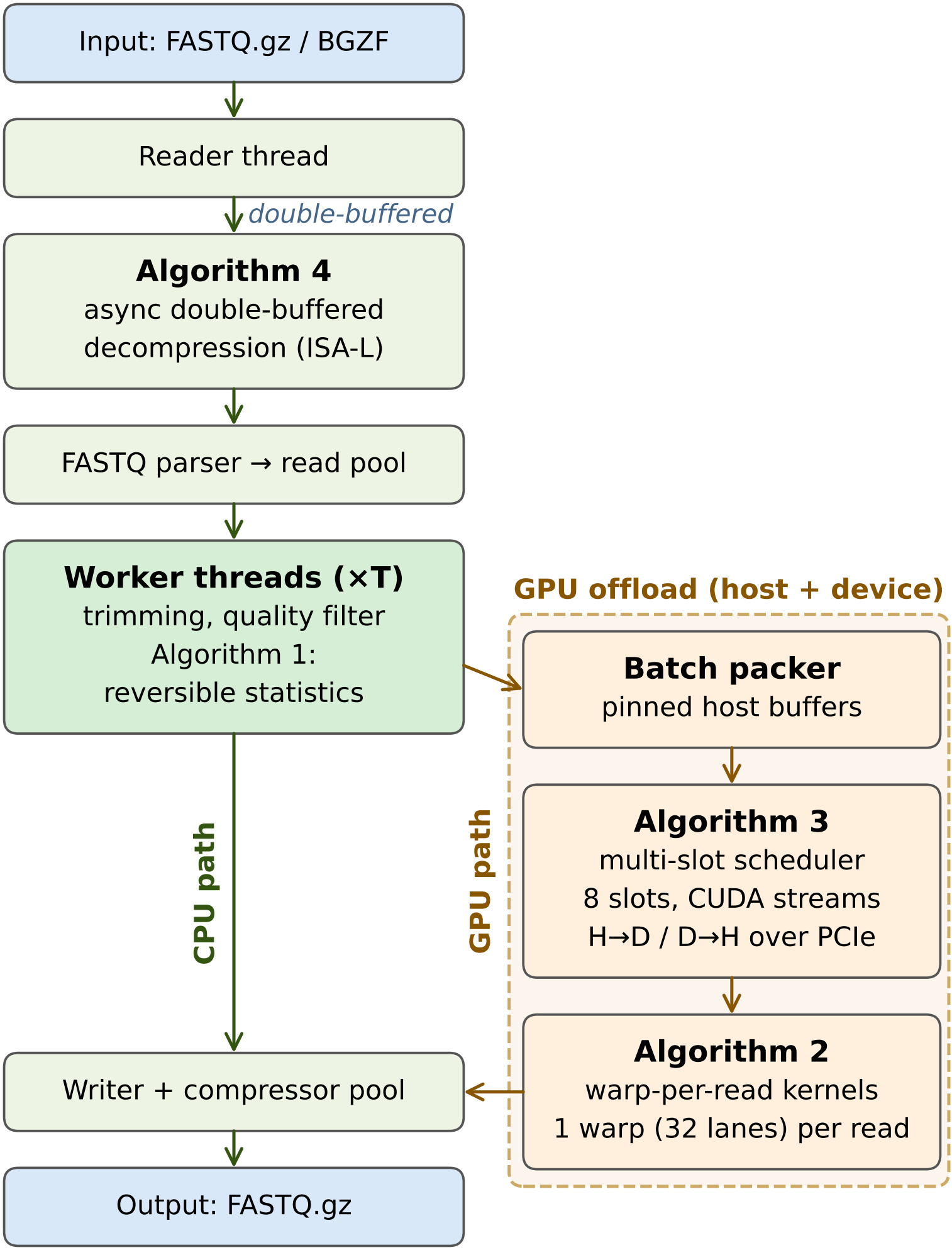
Input: FASTQ.gz / BGZF
Reader thread
double-buffered
Algorithm 4
async double-buffered
decompression (ISA-L)
FASTQ parser → read pool
Worker threads (×T)
trimming, quality filter
Algorithm 1:
reversible statistics
GPU offload (host + device)
Batch packer
pinned host buffers
Algorithm 3
multi-slot scheduler
8 slots, CUDA streams
H→D / D→H over PCIe
Algorithm 2
warp-per-read kernels
1 warp (32 lanes) per read
GPU path
CPU path
Writer + compressor pool
Output: FASTQ.gz